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External member, external member molding method, and electronic instrument

Abstract

A rear panel of an electronic keyboard instrument has a frame portion including multiple sound radiation holes, and the frame portion has a first hole frame including a hole forming surface of the sound radiation hole and a second hole frame including a hole forming surface of the sound radiation hole which is longer in a thickness direction of the frame portion (a Y-axis direction) than the first frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application is based upon and claims the benefit of priority under 35 USC 119 to Japanese Patent Application No. 2021-207675 filed on Dec. 22, 2021, and the content thereof, including the specification, claims, drawings and abstract, is incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

(2) The present disclosure relates to an external member, an external member molding method, and an electronic instrument.

Description of the Related Art

(3) An external member including multiple sound radiation holes like a speaker cover or net has conventionally been used for a sound generating or radiation device such as an electronic instrument or a radio. For example, JP-UM-A-6-62693 discloses a speaker net in which multiple circular sound radiation holes are disposed to be staggered in a zigzag grid-like pattern. This speaker net is attached to a cabinet including a sound hole portion in such a manner as to cover the sound hole portion of the cabinet. A speaker is disposed inside the sound hole portion.

SUMMARY

(4) According to an aspect of the present disclosure, there is provided an external member including a frame portion comprising multiple sound radiation holes, wherein the frame portion includes a first hole frame comprising a hole forming surface of the sound radiation hole, and a second hole frame including a hole forming surface of the sound radiation hole which is longer in a thickness direction of the frame portion than the first hole frame.

(5) According to another aspect of the present disclosure, there is provided an electronic instrument including the external cover described above.

(6) According to a further aspect of the present disclosure, there is provided an external member molding method including clamping a mold made up of a core mold including a first pin which corresponds to the hole forming surface of the first hole frame and a second pin which corresponds to the hole forming surface of the second hole frame, and a cavity mold, and molding the frame portion including the multiple sound radiation holes including the first hole frame and the second hole frame by injecting a molten resin into the mold.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an electronic keyboard instrument including an external member (a rear panel) according to an embodiment or application example of the present disclosure;
- (2) FIG. 2 is a rear view of the electronic keyboard instrument including the external member (the rear panel) according to the application example of the present disclosure;
- (3) FIG. 3 is an enlarged rear view of the electronic keyboard instrument, showing a first sound radiation section and a second sound radiation section which are provided in the external member (the rear panel) according to the application example of the present disclosure;
- (4) FIG. 4 is a sectional view of the external member (the rear panel) according to the application example of the present disclosure taken along a line IV-IV in FIG. 3;
- (5) FIG. 5 is a sectional view of the external member (the rear panel) according to the application example of the present disclosure taken along a line V-V in FIG. 3;
- (6) FIG. 6 is a sectional view of the external member (the rear panel) according to the application example of the present disclosure taken along a line VI-VI in FIG. 3;
- (7) FIG. 7A is an enlarged rear view of a portion P in FIG. 3, which is a partially enlarged view of isomorphic sound radiation portions of the second sound radiation section in the external member (the rear panel) according to the application example of the present disclosure;
- (8) FIG. 7B is a sectional view taken along a line VIIb-VIIb in FIG. 7A, which is a partially enlarged view of the isomorphic sound radiation portions of the second sound radiation section in the external member (the rear panel) according to the application example of the present disclosure;
- (9) FIG. 8 is a sectional view of the external member (the rear panel) according to the application example of the present disclosure taken along a line VIII-VIII in FIG. 3;

(10) FIG. **9** is a perspective view of the second sound radiation section of the external member (the rear panel) according to the application example of the present disclosure as seen from an inner side thereof;

(11) FIG. **10** is a perspective view of the first sound radiation section of the external member (the rear panel) according to the application example of the present disclosure as seen from a front side (an outer side) thereof, in which surfaces to be molded by a core mold are shown as left blank or white, while surfaces to be molded by a cavity mold are shown as shaded;

(12) FIG. **11** is a perspective view of the first sound radiation section of the external member (the rear panel) according to the application example of the present disclosure as seen from a rear side (an inner side) thereof, in which surfaces to be molded by the core mold are shown as left blank or white, while surfaces to be molded by the cavity mold are shown as shaded;

(13) FIG. **12A** is an explanatory diagram of a mold for molding the external member (the rear panel) according to the application example of the present disclosure, which is an enlarged rear view of a front side (an outer side) of the second sound radiation hole in the rear panel, which constitutes a product molded by the mold; and

(14) FIG. **12B** is an explanatory diagram of the mold for molding the external member (the rear panel) according to the application example of the present disclosure, which is a sectional view of the mold corresponding to a sectional view taken along a line XIIb-XIIb in FIG. **12A**.

DESCRIPTION OF THE EMBODIMENT

(15) Hereinafter, referring to accompanying drawings, an embodiment or application example of the present disclosure will be described. An electronic keyboard instrument **10** (an electronic instrument), which is a sound generating device, shown FIG. **1** includes a 61-note keyboard **30** and an instrument case **19**. A control section **31** including an adjustment control **13** is provided on an upper surface **11** of the electronic keyboard instrument **10**. As shown in FIG. **2**, the instrument case **19** includes a rear panel **15**, which is an external member, and a rear cover **12**, which is disposed on an outer side of the rear panel **15**. A connector panel **18**, which includes a connection port for an AC adaptor, connection ports for other sound generating or radiation devices, and USB connection ports, is provided at a central portion of a lower part of the rear panel **15** of the electronic keyboard instrument panel **10**.

(16) In the following description, an up-down direction of the electronic keyboard instrument **10** is referred to as a Z-axis (a lower side is referred to as a positive side of the Z-axis), a left-right direction of the electronic keyboard instrument **10**, which is a direction in which keys are aligned, is referred to as an X-axis (a high-note keys side is referred to as a positive side of the X-axis), and a front-rear direction of the electronic keyboard instrument **10**, which is a front-rear direction of a key, is referred to a Y-axis (a nearer side of the key is referred to as a positive side of the Y-axis).

(17) Two first sound radiation sections **100** and two second sound radiation sections **200** are provided in the rear panel **15** of the electronic keyboard instrument **10**. Each first sound radiation section **100** has a frame portion **110** which includes multiple sound radiation holes **150**, and each second sound radiation section **200** has a frame portion **210** which includes multiple sound radiation holes **250**. The first sound radiation section **100** and the second sound radiation section **200** are disposed in series in a direction (an X-axis direction in FIG. **2**) which is at right angles to an axial direction of the sound radiation holes **150**, **250** (A Y-axis direction in FIG. **2**, that is, a thickness direction of the frame portions **110**, **210**). Then, the two first sound radiation sections **100** are disposed outwards at outer sides (sides farther away from the connector panel **18**) of the electronic keyboard instrument **10** in the X-axis direction, and the two second sound radiation sections **200** are disposed inwards at inner sides (sides closer to the connector panel **18**) of the electronic keyboard instrument **10** in the X-axis direction. The first sound radiation section **100** and the second sound radiation section **200** which make up a pair at a left side and the first sound radiation section **100** and the second sound radiation section **200** which make up a pair at a right side of the electronic keyboard instrument **10** are provided symmetrically with each other.

(18) As shown in FIG. 4, a sound radiation apparatus **20** is provided on an inner side (the positive side of the Y-axis) of the rear panel **15** in such a manner as to correspond individually to the first sound radiation section **100** and the second sound radiation section **200**. In the present application example, the sound radiation apparatus **20** is a speaker. The rear panel **15** (the first sound radiation sections **100**, the second sound radiation sections **200**) is disposed to lie close to the sound radiation apparatuses **20**. The rear cover **12** is provided on the outer side of the rear panel **15**. In the rear cover **12**, portions corresponding to the first sound radiation sections **100** and the second sound radiation sections **200** are opened. Although not shown, a saran net can be provided on an outer surface of the rear cover **12**.

(19) As shown in FIG. 3, the frame portions **110**, **210** are provided in such a manner that an external shape of an outer edge has a substantially oval shape whose major axis extends in the X-axis direction. The frame portions **110**, **210** have multiple frames **115**, **215**, respectively. When referred to here, the frames **115**, **215** denote sides making up, for example, the sound radiation holes **150**, **250**, respectively, which have a hexagonal shape. The sound radiation holes **150**, **250** are provided by areas which are surrounded by the multiple frames **115**, **215**, respectively. In addition, as shown in FIG. 4, surfaces (outer surfaces **110a**, **210a**) of outer sides (a negative side of the Y-axis direction) of the frame portions **110**, **210** are formed flat and level with each other.

(20) The first sound radiation section **100** will be described. As shown in FIG. 3, the sound radiation holes **150** provided in the frame portion **110** of the first sound radiation section **100** include multiple first sound radiation holes **160-1** to multiple sixth sound radiation holes **160-6**, and multiple isomorphic sound radiation holes **170**. The first sound radiation section **100** includes the multiple isomorphic sound radiation holes **170** which are provided on an inner side thereof (a side facing the connector panel **18**) in a direction (the X-axis direction) which is at right angles to the thickness direction (the Y-axis direction) of the frame portion **110**. The frame portion **110** of the first sound radiation section **100** includes straight-line frames **120-1** to **120-6**, which are provided to extend long in a straight line. The straight-line frames **120-1** to **120-4** are provided to extend in the Z-axis direction in parallel to one another in such a manner as to be connected to the outer edge of the frame portion **110**. The straight-line frames **120-1**, **120-2** are provided to be spaced a predetermined distance apart from each other at a substantially central portion of the frame portion **110** in the X-axis direction, and the straight-line frame **120-1** is disposed inwards, while the straight-line frame **120-2** is disposed outwards.

(21) The straight-line frame **120-3** is disposed inwards of the straight-line frame **120-1** in the x-direction, while the straight-line frame **120-4** is disposed outwards of the straight-line frame **120-2** in the X-axis direction. The straight-line frame **120-4** also include a sloping portion (a second sloping portion **162b**) at a portion which is superposed on one of sides of a second sound radiation hole **160-2** which form a hexagonal shape. The straight-line frames **120-5**, **120-6** are provided to extend in the X-axis direction in parallel to each other in such a manner as to be connected to the outer edge of the frame portion **110**. The straight-line frames **120-5**, **120-6** are provided in the vicinity of lower and upper outer edges of the frame portion **110** in the Z-axis direction, respectively in such a manner that the straight-line frame **120-5** lies on a lower side, while the straight-line frame **120-6** lies on an upper side.

(22) The multiple polygonal or substantially hexagonal isomorphic sound radiation holes **170** are provided substantially into a honeycomb configuration at a portion of the frame portion **110** which lies inwards in the X-axis direction with respect to the straight-line frame **120-1**, which is disposed at a central portion of the frame portion **110**, and between the lower and upper straight-line frames **120-5**, **120-6**. As shown in FIG. 4, the isomorphic sound radiation hole **170** is provided to have substantially the same shape at an inner side and an outer side thereof in an axial direction (the Y-axis direction) of the sound radiation hole **150**. Here, when referred to in relation the isomorphic sound holes **170**, **171**, **270**, and the like, the isomorphic sound hole means a hole having substantially the same shape at an inner side and an outer side thereof in the axial direction (the Y-

axis direction, that is, the thickness direction of the frame portion **110**) of the sound radiation hole **150**. The isomorphic sound radiation holes **170** which connect to the outer edge of the frame portion **110** or the straight-line frames **120-5**, **120-6** are provided to have a deformed substantially hexagonal shape or a shape resulting when the substantially hexagonal shape is cut into halves.

(23) On the other hand, multiple polygonal or substantially hexagonal first sound radiation holes **160-1** to third sound radiation holes **160-3** are provided substantially into a honeycomb configuration at a portion of the frame portion **110** which lies outwards in the X-axis direction with respect to the straight-line frame **120-2** and between the straight-line frames **120-5**, **120-6**. The multiple first sound radiation holes **160-1** to third sound radiation holes **160-3** are each aligned in the X-axis direction, while being arranged in three rows in the Z-axis direction. Referring to FIG. 3, the multiple first sound radiation holes **160-1** are aligned horizontally in an upper row, the multiple second sound radiation holes **160-2** are aligned horizontally in a middle row, and the multiple third sound radiation holes **160-3** are aligned horizontally in a lower row. Here, in FIG. 3, sloping surfaces of the first sound radiation holes **160-1** to seventh sound radiation holes **160-7** are shown as being shaded.

(24) The first sound radiation hole **160-1** will be described by taking a first sound radiation hole **161** in the first sound radiation holes **160-1** which is shown in a sectional view in FIG. 5 for example. The first sound radiation hole **161** has a first sloping portion **161b** including a first sloping surface **161a** which is disposed to slope down from the thickness direction of the frame portion **110** which is an axial direction of the first sound radiation hole **161** (the Y-axis direction, that is, the thickness direction of the frame portion **110**) within an area in the first sound radiation hole **161** (within an area inside the substantially hexagonal shape when seen from above in FIG. 3). A first direction **161c**, which is a sloping direction of the first sloping surface **161a** (a direction at right angles to an edge line **161a1** of the first sloping surface **161a** inside the hole, which is a direction directed from an inner side to an outer side of the first sound radiation hole **161**, that is, a sound radiation direction), is a direction which divides an angle formed between a negative side of the X-axis and a negative side of the Z-axis substantially into half angles.

(25) Similarly, a second sound radiation hole **160-2** will be described by taking a second sound radiation hole **162** shown in the sectional view in FIG. 4 for example. The second sound radiation hole **162** has a second sloping portion **162b** including a second sloping surface **162a** which is disposed to slope down from the thickness direction of the frame portion **110** which is an axial direction of the second sound radiation hole **162** (the Y-axis direction) within an area in the second sound radiation hole **162**. Then, a second direction **162c**, which is a sloping direction of the second sloping surface **162a**, constitutes a direction of the negative side of the X-axis.

(26) Also, similarly, a third sound radiation hole **160-3** (a third sound radiation hole **163**) has a third sloping portion **163b** including a third sloping surface **163a**, and a third direction **163c**, which is a sloping direction of the third sloping surface **163a**, constitutes a direction which divides an angle formed between the negative side of the X-axis and a positive side of the Z-axis substantially into half angles. The first direction **161c**, the second direction **162c**, and the third direction **163c** constitute different directions from one another.

(27) On the other hand, multiple quadrangular fourth sound radiation holes **160-4** and multiple quadrangular fifth sound radiation holes **160-5** are disposed on outer sides of the upper straight-line frame **120-6** and the lower straight-line frame **120-5**, respectively. In the fourth sound radiation hole **160-4** (a fourth sound radiation hole **164**), a fourth direction **164c**, which is a direction of the positive side of the Z-axis, constitutes a sloping direction of a fourth sloping surface **164a** of a fourth sloping portion **164b**. In the fifth sound radiation hole **160-5** (a fifth sound radiation hole **165**), a fifth direction **165c**, which is a direction of the negative side of the Z-axis, constitutes a sloping direction of a fifth sloping surface **165a** of a fifth sloping portion **165b**. The fourth sloping portion **164b** and the fifth sloping portion **165b** include the sloping surfaces (the fourth sloping surface **164a**, the fifth sloping surface **165a**) which are both disposed to slope down from the

thickness direction of the frame portion **110**.

(28) Also, on the other hand, an isomorphic sound radiation hole **172** is provided substantially at a central portion of the frame portion **110** which is defined between the central straight-line frames **120-1**, **120-2** in the Z-axis direction and the X-axis direction. The sixth sound radiation hole **160-6** and the seventh sound radiation hole **160-7**, which each have a substantially rectangular shape which is elongated in the Z-axis direction, are provided at an upper side (the negative side) and a lower side (the positive side) of the isomorphic sound radiation hole **172** in the Z-axis direction, respectively.

(29) As shown in FIG. **6**, a sixth sloping surface **166a** of a sixth sloping portion **166b** of the sixth sound radiation hole **160-6** is disposed to slope down from the thickness direction of the frame portion **110**. In other words, the sixth sloping surface **166a** is provided in such a manner as to expand from the inner side towards the outer side of the frame portion **110**. The sixth sloping portion **166b** includes a bent portion which is provided in such a manner as to be bent in the expanding direction. Similarly, a seventh sloping surface **167a** of a seventh sloping portion **167b** of the seventh sound radiation hole **160-7** includes a bent portion, which is similar to that described above, and is disposed to slope down from the thickness direction of the frame portion **110**. A sloping direction of the sixth sloping surface **166a** of the sixth sound radiation hole **160-6** is the same as the fourth direction **164c**. A sloping direction of the seventh sloping surface **167a** of the seventh sound radiation hole **160-7** is the same as the fifth direction **165c**. The sloping portions (the sixth sloping portion **166b** and the seventh sloping portion **167b**) which include the sloping surfaces (the sixth sloping surface **166a** and the seventh sloping surface **167a**) are both provided between second hole frames **112**, which will be described later. Then, the sloping portions (the sixth sloping portion **166b** and the seventh sloping portion **167b**) are provided to make a pair so that the sloping surfaces (the sixth sloping surface **166a** and the seventh sloping surface **167a**) face each other.

(30) In this way, as shown in FIG. **3**, the sound radiation directions from the first sound radiation section **100** include a right-upward direction (the first direction **161c**) defined by the first sound radiation hole **160-1**, a rightward direction (the second direction **162c**) defined by the second sound radiation hole **160-2**, a right-downward direction (the third direction **163c**) defined by the third sound radiation hole **160-3**, an upward direction (the fourth direction **164c**) defined by the fourth sound radiation hole **160-4** and the sixth sound radiation hole **160-6**, and a downward direction (the fifth direction **165c**) defined by the fifth sound radiation hole **160-5** and the seventh sound radiation hole **160-7**. As a result, sounds radiated from the first sound radiation section **100** can spread widely.

(31) Also, here, heights (length in the thickness direction of the frame portion **110** (the Y-axis direction)) of the multiple frames **115** which make up the frame portion **110** of the first sound radiation section **100** will be considered. This will be described by taking isomorphic sound radiation holes **171**, **172** shown in FIGS. **7A** and **7B** for example. The multiple frames **115** which make up the frame portion **110** of the first sound radiation section **100** include first hole frames **111** and second hole frames **112** which include hole forming surfaces which define a sound radiation hole **150**. As shown in FIG. **7A**, which is an enlarged view of a portion P in FIG. **3**, an isomorphic sound radiation hole **171** is made up of respective hole forming surfaces **111a** of six first hole frames **111** (denoted by reference signs a to f). Similarly, an isomorphic sound radiation hole **172** is made up of respective hole forming surfaces **112a** of six second hole frames **112** (denoted by reference signs g to l). Here, the hole forming surfaces **111a**, **112a** are side surfaces of a hole which extends in the thickness direction of the frame portion **110** (the Y-axis direction). As a result, a length of the hole forming surface is considered based on the thickness direction of the frame portion **110**, even though the hole forming surface is shaped to expand as it extends.

(32) Then, as shown in FIG. **7B**, a length T2 of the hole forming surface **112a** of the second hole frame **112** in the Y-axis direction of the isomorphic sound radiation hole **172** (the thickness

direction of the frame portion **110**) is longer than a length **T1** of the hole forming surface **111a** of the first hole frame **111** of the isomorphic sound radiation hole **111a** in the thickness direction of the frame portion **110**. Since the outer surface **110a** of the frame portion **110** is formed into a flat plane, an inner surface **110b** of a central portion of the frame portion **110** is made to protrude at the location of the second hole frame **112**. As shown in FIG. 4, the second hole frame **112** can be provided in such a manner as to correspond to a central recessed portion of a speaker cone **21** of the sound radiation apparatus **20**. Here, the second hole frame **112** includes a portion which is situated in an interior portion of the speaker cone **21**. The first hole frame **111** is not situated in the interior portion of the speaker cone **21**.

(33) Subsequently, the second sound radiation section **200** will be described. Sound radiation holes **250** provided in the frame portion **210** of the second sound radiation section **200** are made up of multiple isomorphic sound radiation holes **270**. In other words, the second sound radiation section **200** includes the frame portion **210** having sound radiation holes **250** which are made up of the isomorphic sound radiation holes **270** in place of the multiple sound radiation holes **150** including the first sound radiation hole **160-1** to the seventh sound radiation hole **160-7** of the first sound radiation section **100**. The multiple isomorphic sound radiation holes **270** each have a polygonal or hexagonal shape and are formed substantially into a honeycomb configuration. Additionally, as with the first sound radiation section **100**, straight-line frames **220-1** to **220-6** are provided in the frame portion **210** of the second sound radiation section **200**.

(34) Next, as shown in FIG. 9, frames **215** of the second sound radiation section **200** include first frames **211** and second frames **212** which include hole forming surfaces **211a**, **212a** of sound radiation holes **250**, respectively. As shown in FIG. 8, a length **T4** of the hole forming surface **212a** of the second hole frame **212** in an axial direction of the sound radiation hole **250** (the Y-axis direction, a thickness direction of the frame portion **210**) is longer than a length **T3** of the hole forming surface **211a** of the first hole frame **211** in the thickness direction of the frame portion **210** (the Y-axis direction). In addition, the length **T4** is shorter than the length **T2**.

(35) Since the second hole frame **112** is given the protruding shape, when seen from the side, with a view to spreading sounds as by the sixth sound radiation hole **160-6** and the seventh sound radiation hole **160-7**, the sloping surface is provided long, which results in the fact that the length **T2** is made relatively long. On the other hand, since the frame **215** of the second sound radiation section **200** does not have to spread sounds from the viewpoint of sound characteristics, the length **T4** is made shorter than the length **T2**. The disposition of the frame portions **110**, **210** is not limited to the application example, and hence, the frame portions **110**, **210** can be disposed differently and the lengths **T2**, **T4** can also be changed in accordance with a desired sound characteristic.

(36) In the second sound radiation section **200**, a predetermined area **Q** lying substantially at a central portion of the frame portion **210** is made up of the second hole frames **212** so as to increase the strength of the frame portion **210**. In addition, as shown in FIG. 4, the second hole frames **212** can be provided in such a manner as to correspond to the central recessed portion of the speaker cone **21**.

(37) Here, the rear panel **15** is molded from a resin through injection molding. As this occurs, a draft expanding from the outer side towards the inner side of the rear panel **15** is provided on hole forming surfaces of the sound radiation holes **150**, **250**. For example, as shown in FIGS. 7B and 8, a draft is provided on the hole forming surfaces **111a**, **112a**, **211a**, **212a** as a result of the hole forming surfaces **111a**, **112a**, **211a**, **212a** expanding slightly from the outer side towards the inner side of the rear panel **15**.

(38) A mold **50** for molding the rear panel **15** has a cavity mold **51** and a core mold **52**. In FIGS. **10**, **11** showing the first sound radiation section **100**, shaded portions (a front side of the rear panel **15**) constitute surfaces to be molded by the cavity mold **51**, while blank or white portions constitute surfaces to be molded by the core mold **52**. Four bosses **15a** for attachment of the rear cover **12** are provided on an outer circumference of a front side (an outer side) of the first sound radiation

section **100**. Then, as shown in FIG. **11**, a boss **15b** for connection with the instrument case of the electronic keyboard instrument **10** is provided on a circumference of a rear side (an inner side) of the first sound radiation section **100**.

(39) In molding hexagonal holes in the isomorphic sound radiation holes **170**, **270**, hexagonal holes can be molded by providing hexagonal prism-like projections on the core mold **52**. In molding the first sound radiation hole **160-1** to the third sound radiation hole **160-3**, since the sloping portions (the first sloping portion **161b** to the third sloping portion **163b**) are provided in the first to third sound radiation holes **160-1** to **160-3**, the cavity mold **51** and the core mold **52** are configured as below.

(40) Hexagonal prisms **53** including corresponding sloping surfaces **51a**, **52a** which slope in the same direction on facing side surfaces thereof are provided on the mold **50** in such a manner that a pin **51b**, which constitutes one side or part of the hexagonal prism **53** when the hexagonal prism **53** is divided along a line connecting facing apex portions **M1**, **M2** thereof, is provided on the cavity mold **51** and a pin **52b**, which is made up of the other side or part of the hexagonal prism **53** so divided, is provided on the core mold **52**.

(41) A molding method of the rear panel **15**, which constitutes the external member, using this mold **50** includes clamping the mold **50** and injecting a molten resin into the mold **50** so clamped. Then, a frame portion **110** including a first sound radiation hole **160-1** to a third sound radiation hole **160-3** in which sloping surfaces (a first sloping surface **161a** and the like) are provided within hole areas thereof is molded by the hexagonal prisms **53**.

(42) In this way, in the cavity mold **51** and the core mold **52** for molding the first sound radiation hole **160-1** to the third sound radiation hole **160-3**, the line connecting the facing apex portions **M1**, **M2** of the hexagonal shape constitutes a parting line **PL**.

(43) Further, a molding method of the first hole frames **111**, **211** and the second hole frames **112**, **212** will be described using FIGS. **7A**, **7B**, which show a circumferential area of the isomorphic sound radiation holes **171**, **172** of the first sound radiation section **100**. As shown in FIG. **7B**, the core mold **52** of the mold **50** includes the first pin **52c** corresponding to the hole forming surface **111a** of the first hole frame **111** and the second pin **52d** corresponding to the hole forming surface **112a** of the second hole frame **112**. Then, the molding method of the rear panel **15** using the mold **50** similarly includes clamping the mold **50** and injecting a molten resin thereinto. Then, a frame portion **110** is molded which includes multiple sound radiation holes **150** including first hole frames **111** and second hole frames **112**.

(44) While FIGS. **7A** and **7B** illustrate the first sound radiation section **100**, what is illustrated therein can also equally be applied to the first hole frames **211** and the second hole frames **212** of the second sound radiation section **200**.

(45) Thus, as has been described heretofore, in the application example of the present disclosure, the rear panel **15**, which is the external member, of the electronic keyboard instrument **10** has the frame portions **110**, **210** which include the multiple sound radiation holes **150**, **250**, respectively, and the frame portions **110**, **210** have the first hole frames **111**, **211**, respectively, which include, respectively, the hole forming surfaces **111a**, **211a** of the sound radiation holes **150**, **250** and the second hole frames **112**, **212** which include, respectively, the hole forming surfaces **111a**, **211a** of the sound radiation holes **150**, **250** which are longer in the thickness direction of the frame portions **110**, **210** (the Y-axis direction) than the first hole frames **111**, **211**.

(46) As a result, the sound radiation holes **150**, **250** which are provided by the second hole frames **112**, **212** can be formed into a cylindrical shape which is longer than the sound radiation holes **150**, **250** which are provided by the first hole frames **111**, **211**. Thus, the strength of the frame portions **110**, **210** can be increased even partially while increasing the directivity of sounds radiated from the sound radiation apparatus **20**. Here, although it is also considered that the lengths of all the sound radiation holes **150**, **250** are elongated in the axial direction thereof, in the event that this configuration is adopted, the size of the electronic keyboard instrument **10** to which the external

member is attached will be increased accordingly.

(47) In addition, the second hole frames **112**, **212** are disposed in the central portions of the frame portions **110**, **210**, respectively. As a result, the cylindrical sound radiation holes **150**, **250** which are made up of the second frames **112**, **212** can easily be disposed at the center of sounds radiated from the sound radiation apparatus **20**.

(48) The sloping portions (the sixth sloping portion **166b** and the seventh sloping portion **167b**) including, respectively, the sloping surfaces (the sixth sloping surfaces **166a** and the seventh sloping surfaces **167a**) which are disposed to slope from the thickness direction of the frame portion **110** (the Y-axis direction) are provided between the second hole frames **112**. As a result, the sound radiation directions of sounds can arbitrarily be set while increasing the directivity of the sounds.

(49) Additionally, the sloping portions (the sixth sloping portion **166b** and the seventh sloping portion **167b**) are provided to make a pair so that the sloping surfaces (the sixth sloping surfaces **166a** and the seventh sloping surfaces **167a**) face each other. As a result, the space between the sloping surfaces (the sixth sloping surfaces **166a** and the seventh sloping surfaces **167a**) on the inner side of the rear panel **15**, which is the external member, is narrowed, thereby making it possible to radiate sounds by increasing the sound pressure.

(50) The frame portions **110**, **210** include, respectively, the straight-line frames **120-1** to **120-6**, **220-1** to **220-6**. As a result, the strength of the frame portions **110**, **210** can be increased further.

(51) The frame portions **110**, **210** are provided in such a manner that the outer surfaces (the outer surfaces **110a**, **210a**) are flat and level with each other. As a result, the inner sides of the frame portions **110**, **210** are made to protrude, whereby the external appearance of the rear panel **15** is improved, and a cloth member such as a saran net can also be easily attached to the rear panel **15**.

(52) The speakers, which are the sound radiation apparatuses **20**, are provided in the positions corresponding to the second hole frames **112**, **212**. The second hole frames **112**, **212**, which make up the cylindrical sound radiation holes, enable the speakers to radiate sounds whose directivity is increased.

(53) The part of the inside of the second hole frame **112** is situated in the interior portion of the speaker cone **21** of the speaker. Since only the portion of the frame portion **110** which corresponds to the recessed portion of the speaker cone **21** enters the interior portion of the speaker cone **21**, the frame portion **110** is made to hardly contact the speaker cone **21** even though the speaker cone **21** vibrates. In addition, since the frame portion **110** (the rear panel **15**) can be disposed close to the speaker cone **21**, the electronic keyboard instrument **10** can be made small in size.

(54) The first sound radiation section **100** and the second sound radiation section **200** have the frame portion **110** and the frame portion **210**, respectively. In addition, the length T2 of the hole forming surface **112a**, which makes up the isomorphic sound radiation hole **172** (the sound radiation hole **150**), of the second hole frame **112** of the first sound radiation section **100** which is defined in the thickness direction of the frame portion **110** is longer than the length T4 of the hole forming surface **212a**, which makes up the sound radiation hole **250**, of the second frame **212** of the second sound radiation section **200** which is defined in the thickness direction of the frame portion **210**. As a result, the lengths T2, T4 of the hole forming surfaces of the first hole frames **111**, **211** and the second hole frames **112**, **212** can be set in accordance with the desired sound characteristics.

(55) The rear panel **15**, which is the external member, is provided on the electronic keyboard instrument **10**, which is an electronic instrument. As a result, the electronic instrument can be provided which includes the external member whose strength is improved while increasing the sound directivity.

(56) The molding method of the rear panel **15** includes clamping the mold **50** having the core mold **52** including the first pin **52c** which corresponds to the hole forming surfaces **111a**, **211a** of the first hole frames **111**, **211** and the second pin **52d** which corresponds to the hole forming surfaces **112a**,

212a of the second hole frames **112**, **212**, and the cavity mold **52**, and molding the frame portions **110**, **210** which include the multiple sound radiation holes **150**, **250** which include the first hole frames **111**, **211** and the second hole frames **112**, **212** by injecting a molten resin into the mold **50**. As a result, the molding method of the rear panel **15** can be provided in which the strength is improved while increasing the sound directivity.

(57) The drafts on the hole forming surfaces **111a**, **211a** of the first hole frames **111**, **211** and the hole forming surfaces **112a**, **212a** of the second hole frames **112**, **212** are provided in such a manner as to expand from the outer side towards the inner side of the frame portions **110**, **210**, respectively. As a result, the molding method of the rear panel **15** can be provided in which the sound radiation properties are increased by increasing the sound pressure, particularly, in the isomorphic sound radiation holes **170**, **270**.

(58) While the application example has been described heretofore, the application example is presented as the example, and hence, there is no intention to limit the scope of the present invention by the example. The novel application example can be carried out in other various forms, and various omissions, replacements and modifications can be made thereto without departing from the spirit and scope of the present invention. Those resulting application examples and modified examples thereof are included in the scope and gist of the present invention and are also included in the scope of inventions claimed for patent under claims below and their equivalents.

Claims

1. An external member comprising: a frame portion comprising multiple sound radiation holes, wherein the frame portion comprises: first hole frames each comprising a hole forming surface of a respective one of the sound radiation holes; and second hole frames each comprising a hole forming surface of a respective one of the sound radiation hole which is holes, the second hole frames being longer in a thickness direction of the frame portion than the first hole frame frames, and wherein a sloping portion including a sloping surface which is disposed to slope along the thickness direction of the frame portion is provided between the second hole frames.
2. The external member according to claim 1, wherein the second hole frames are disposed at a central portion of the frame portion.
3. The external member according to claim 1, wherein the sloping portion comprises a pair of sloping surfaces provided so as to face each other.
4. The external member according to claim 1, wherein the frame portion comprises a straight-line frame.
5. The external member according to claim 1, wherein an outer surface of the frame portion is flat and level.
6. The external member according to claim 1, wherein the second hole frame is provided at a position corresponding to a speaker.
7. The external member according to claim 6, wherein an inner portion of the second hole frame is positioned in an interior portion of a speaker cone of the speaker.
8. An external member comprising: a first sound radiation section; and a second sound radiation section, wherein each of the first sound radiation section and the second sound radiation section comprises a frame portion comprising multiple sound radiation holes, the frame portion comprising: first hole frames each comprising a hole forming surface of a respective one of the sound radiation holes; and second hole frames each corresponding to a hole forming surface of a respective one of the sound radiation holes, the second hole frames being longer in a thickness direction of the frame portion than the first hole frames, and wherein a length of the hole forming surfaces of the second hole frames of the first sound radiation section in the thickness direction of the frame portion is longer than a length of the hole forming surfaces of the second hole frames of the second sound radiation section in the thickness direction of the frame portion.

9. The external member according to claim 8, wherein the second hole frames are disposed at a central portion of the frame portion.

10. The external member according to claim 8, wherein a sloping portion including a sloping surface which is disposed to slope along the thickness direction of the frame portion is provided between the second hole frames.

11. An electronic instrument comprising the external member according to claim 1.

12. An external member molding method comprising: clamping a mold comprising a core mold comprising a first pin which corresponds to a hole forming surface of a first hole frame and a second pin which corresponds to a hole forming surface of a second hole frame, and a cavity mold; and molding a frame portion comprising multiple sound radiation holes including the first hole frame and the second hole frame by injecting a molten resin into the mold, wherein drafts on the hole forming surface of the first hole frame and the hole forming surface of the second hole frame are provided in such a manner as to expand from an outer side towards an inner side of the frame portion.
